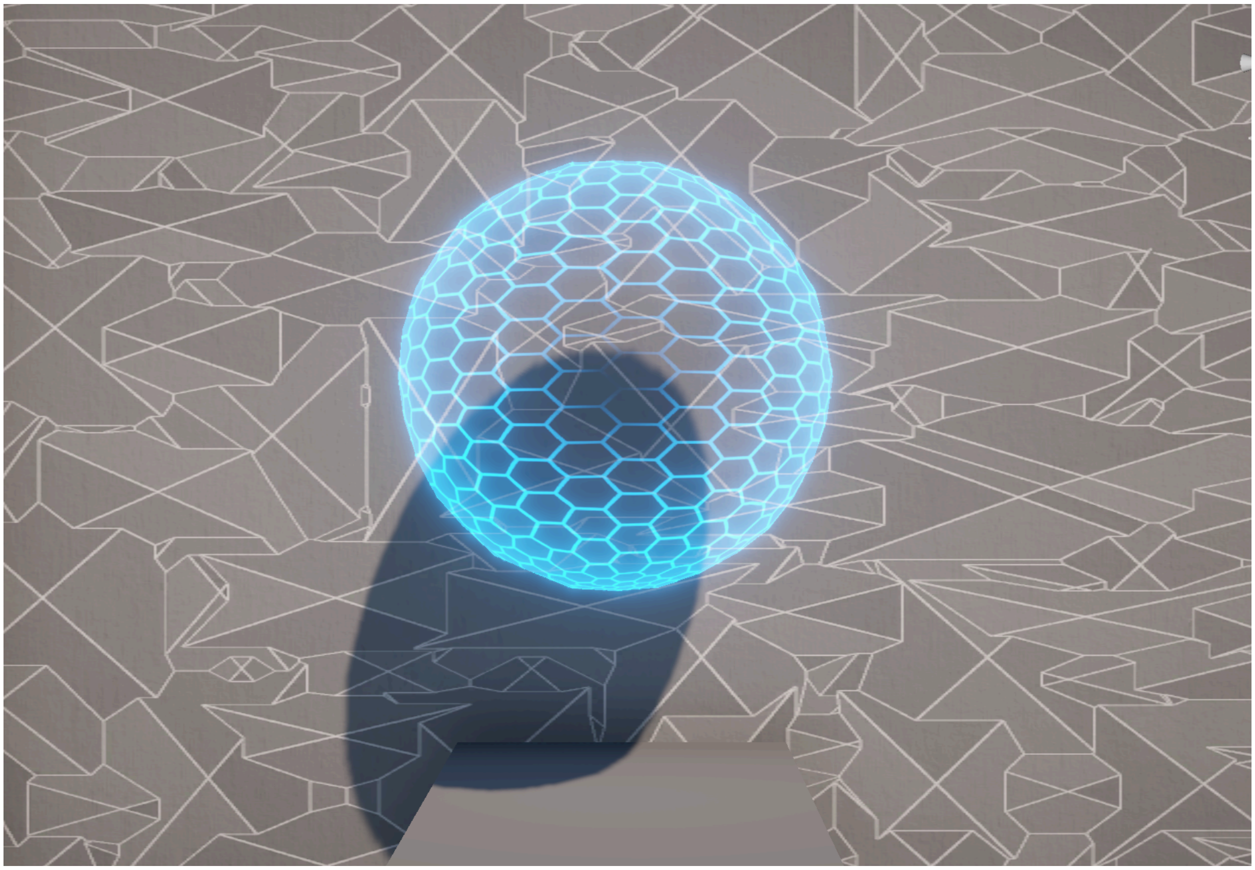
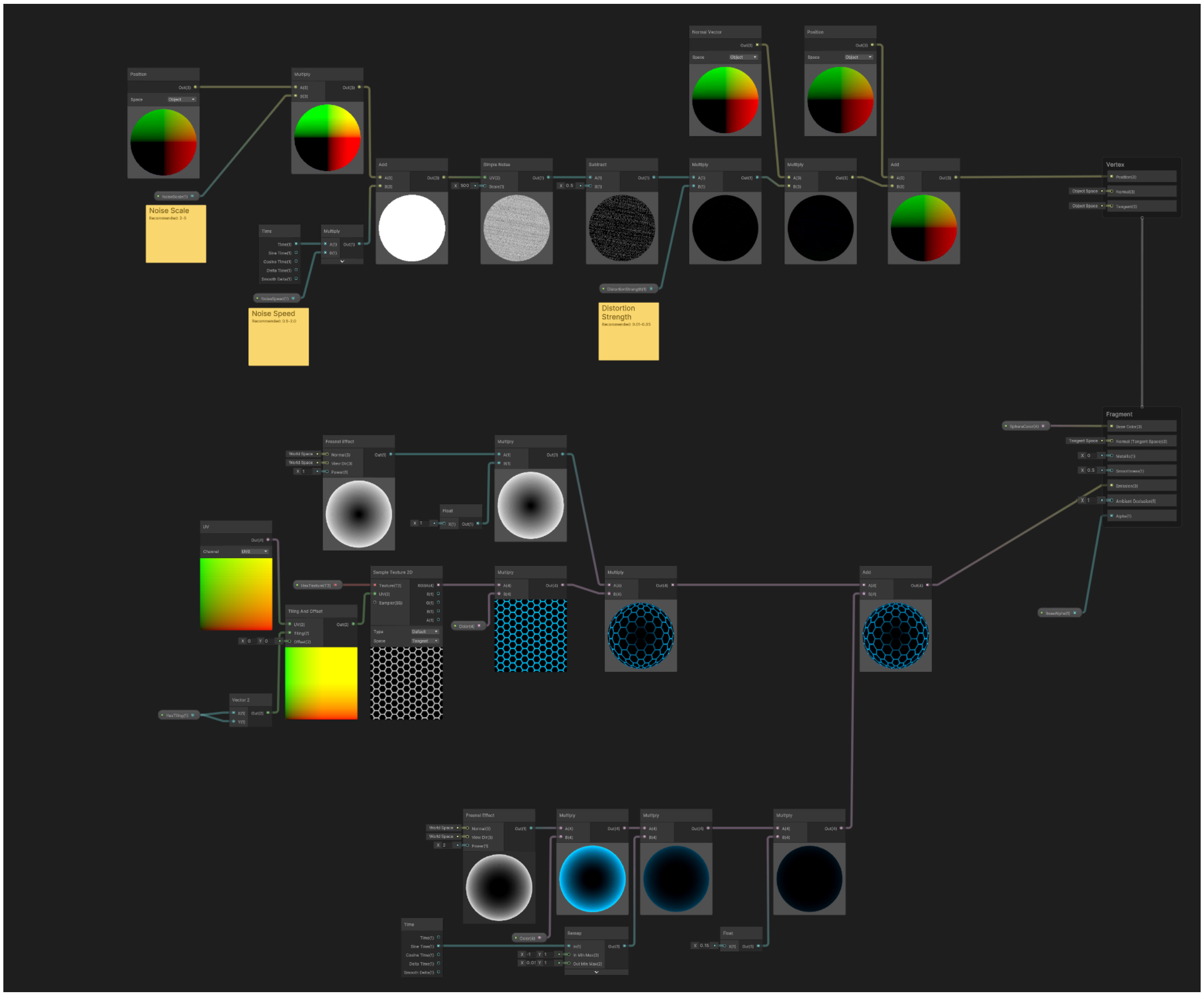
Energy Shield - Unity Shader Graph



This shader creates a translucent hexagonal grid around an object, similar to an energy shield. The shield has a glowing effect around the edges and is more transparent at the core.

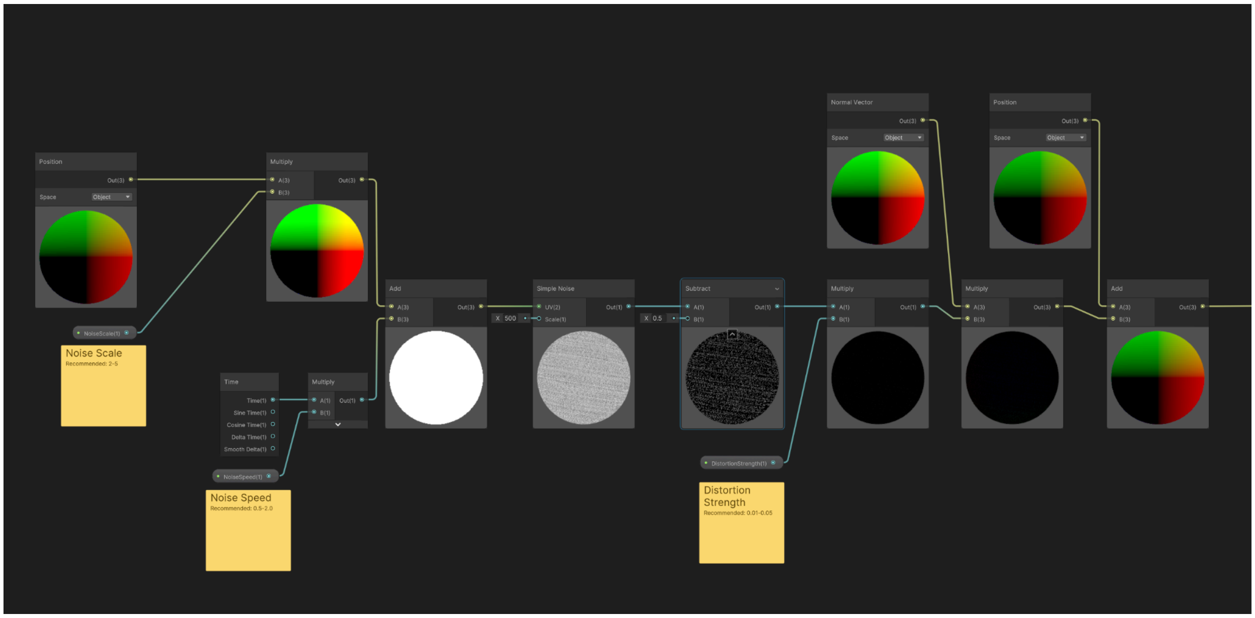
Here is the layout of the shader:



There are 3 main components:

- Distortion (Top)
- Hexagons (Middle)
- Outer Glow (Bottom)

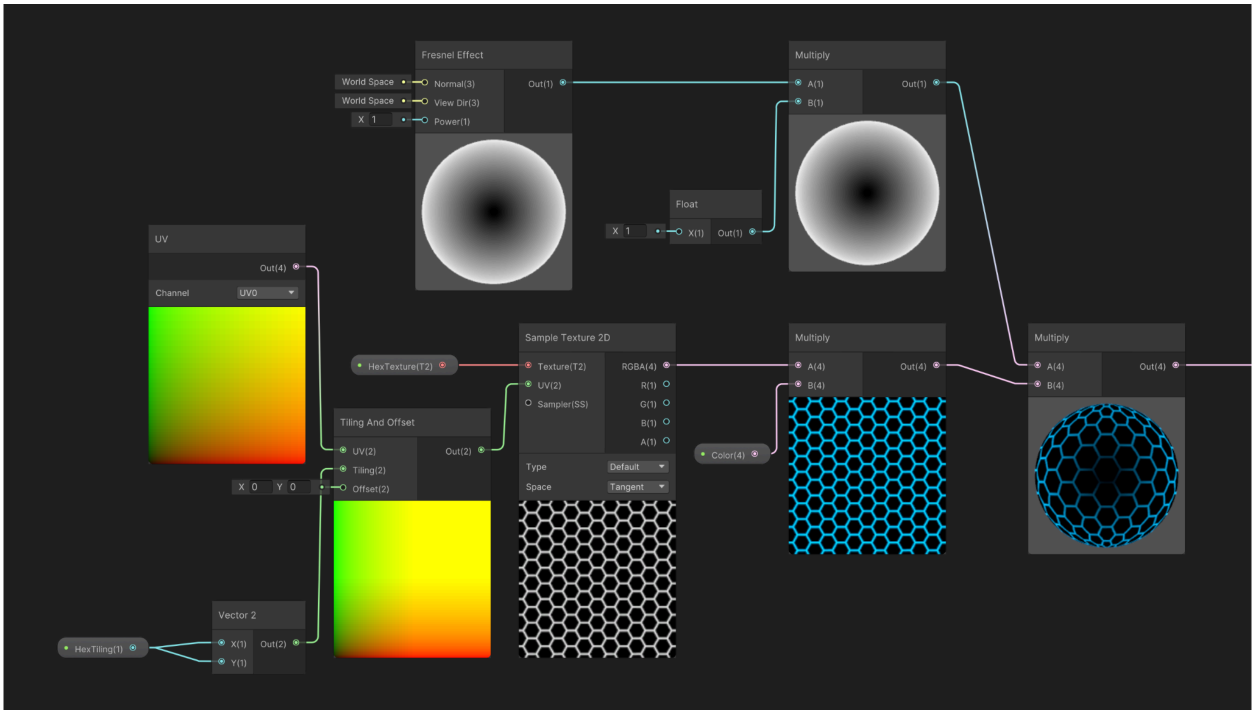
First, we can examine the distortion layer:



We start by making a Position Node and setting it to Object Space. We then multiply it by our NoiseScale (recommended value: 2-5). Then, add a Time Node and multiply our NoiseSpeed. Add (time * speed) + (Position * scale). The result will determine the movement of our noise.

Take the resulting Add Node and pass it as the UV to a Simple Noise Node. Then subtract 0.5 to create an inversion. Multiply the result with a float for Distortion Strength (recommended value: 0.01-0.05). Multiply the result of that with a Normal Vector Node, then Add that to another Object Space Position Node.

Next, we can view the hexagon pattern shader:

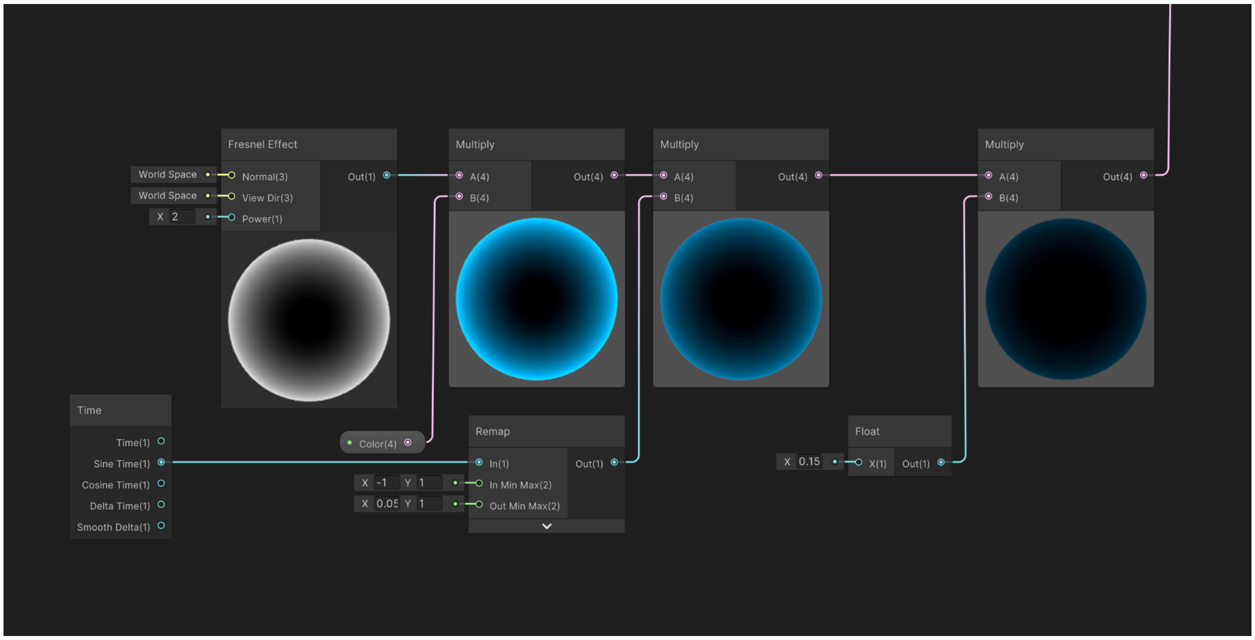


I wanted the hex tiling to be the same on the X and Y axes, so I have one float for scale that I expand into a Vector2. I set HexTiling to 4. Higher values create smaller hexagons. Create a Tiling and Offset Node. Create a UV Node for the UV, and pass the Tiling vector into the Tiling input.

In order for this effect to work, you need a black and white hexagon pattern. Once you find one, create a texture for it, then make a Sample Texture 2D node, with the hexagons as the texture, and the Tiling And Offset as the UV. Multiply it with your color of choice.

Then create a Fresnel Effect Node. This makes a white glow on the outskirts of the material. You can optionally choose to multiply the Node with a float to make the effect brighter. I left it as 1 here. Multiply the Fresnel effect with the grid. This creates the hexagon pattern with reduced opacity in the center.

Finally, let's examine the outer glow:



This one is pretty simple. We use a Fresnel Effect Node once again. Create a Fresnel Node and multiply it with your color of choice. Create a Time node, and pass Sine Time into a Remap Node. Map in (-1, 1) and Out Min-Max (0.05, 1). Then multiply (remapped time * colored fresnel). This produces a particularly bright outer color, so you can choose to multiply this with a float to reduce or strengthen the effect.

Add the hexagon pattern to the colored fresnel node, then map that final Add Node to the Emission property of the shader.

Take the Distortion effect, and map it to the Vertex Position. This will cause the intended jitters in the edges of the surface.

Create a BaseAlpha variable and set it to a low value like 0.2. Map that to the alpha of the material.